



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO SELVI AGENCY

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

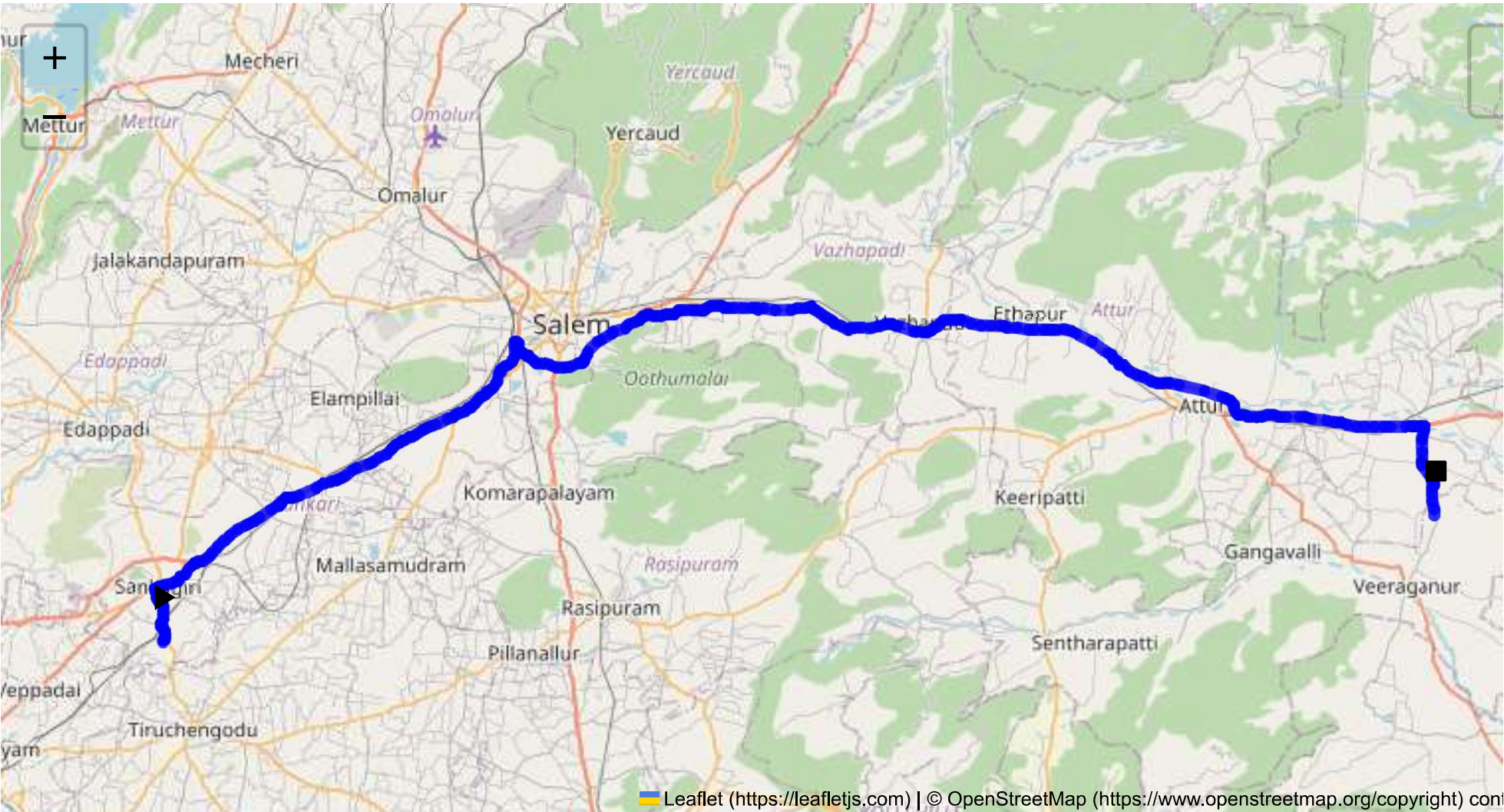
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 120.41 km**  
**Estimated Duration: 2.3 hours**  
**Adjusted Duration (Heavy Vehicle): 2.8 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.5248, 78.7572)**



**Welcome to the Journey Risk Management Study**

**Route Safety Analysis**

**1. Overview of the Route Map**



The route from Sangagiri, Tamil Nadu to Navalur, Tamil Nadu, covers approximately 120.41 kilometers. It primarily follows state highways and local roads, connecting rural and semi-urban areas. The journey usually takes around 2.26 hours for heavy vehicles. The drive encompasses a mix of flat terrains and minor elevation changes, with agricultural lands and small towns.

## 2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather:** Tamil Nadu typically experiences a tropical climate with high temperatures and humidity. The southwest (June to September) and northeast (October to December) monsoons bring heavy rainfall that may cause flooding and road blockage.
- **Hazards:** During monsoon seasons, the risk of slippery roads and reduced visibility is high. Waterlogging and landslides on certain stretches can occur, particularly in lower-lying areas.

## 3. Analysis of Traffic Patterns

- **Peak Hours:** Traffic congestion is common during morning (7-9 AM) and evening (5-7 PM) in towns like Salem and smaller junctions.
- **Congestion Areas:** Bottlenecks typically occur at intersections linking major highways and local roads, especially near Salem, and small towns en route, due to mixed traffic including public transport and commercial vehicles.

## 4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Varies significantly along the route; highways generally have good-quality pavement, but local roads may suffer from potholes and uneven surfaces.
- **Infrastructure:** Adequate signage and lighting are present on highways, but may be deficient on rural roads. Some stretches may lack hard shoulders and adequate barricades.

## 5. Suggestions for Alternative Routes for Emergencies

In case of emergencies or obstructions, alternative routing through NH544 and NH79 may provide detours around congested or unsafe segments, although a longer distance may be involved.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

- Transport of hazardous materials requires adherence to specific permits and safety guidelines as per Tamil Nadu and central Indian laws.
- Use of designated routes, timing restrictions, and safety bottlenecks must be followed to minimize risk.

## 7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

Previous incidents in the region often relate to over-speeding and vehicular malfunction. Although rare, hazardous material spillages near farmland can have severe environmental consequences.

8. Environmental Considerations and Sensitive Areas

- Important zones include agricultural lands and small water bodies, which can be affected by spillage or pollution.
- Areas near large green cover or water catchment areas should be approached cautiously to prevent ecological disruption.

9. Analysis of Communication Coverage

- Mobile coverage is generally reliable on highways and near towns. However, rural stretches may experience short dead zones affecting communication.
- Drivers are advised to be prepared with offline navigation and emergency contact information.

10. Estimated Emergency Response Times

- Urban areas like Salem typically have faster emergency response times (15-30 minutes).
- Remote areas may experience delayed responses due to distance and lesser infrastructure support (up to 1 hour or more).

11. Overall Summary of Risk Assessment

The route features a mix of urban and rural challenges. Monsoon weather poses significant risks, particularly relating to road conditions and visibility. While most of the route is covered by reliable communication networks, certain zones may face connectivity issues, affecting timely support in emergencies. Compliance with transport regulations for hazardous materials is crucial to preventing incidents. Overall, while navigable, this route requires careful planning and alertness, especially concerning road quality and weather conditions.

Risk Assessment - Turns

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Roundabout | High       | 11.63303, 78.12713 | 15 KM/Hr    |
| 1 | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 2 | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 3 | Turn       | Medium     | 11.44900, 77.87406 | 30 KM/Hr    |
| 4 | Turn       | Medium     | 11.45000, 77.87213 | 30 KM/Hr    |
| 5 | Turn       | High       | 11.45028, 77.87202 | 15 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 6  | Turn      | Medium     | 11.45309, 77.87325 | 30 KM/Hr    |
| 7  | Turn      | Medium     | 11.46400, 77.87323 | 30 KM/Hr    |
| 8  | Turn      | Medium     | 11.46765, 77.87037 | 30 KM/Hr    |
| 9  | Turn      | High       | 11.47403, 77.86841 | 15 KM/Hr    |
| 10 | Turn      | High       | 11.49149, 77.89601 | 15 KM/Hr    |
| 11 | Turn      | High       | 11.49224, 77.89601 | 15 KM/Hr    |
| 12 | Turn      | Medium     | 11.53501, 77.95707 | 30 KM/Hr    |
| 13 | Turn      | Medium     | 11.53601, 77.95856 | 30 KM/Hr    |
| 14 | Turn      | Medium     | 11.64172, 78.11817 | 30 KM/Hr    |
| 15 | Turn      | Medium     | 11.64216, 78.11875 | 30 KM/Hr    |
| 16 | Turn      | High       | 11.64195, 78.12129 | 15 KM/Hr    |
| 17 | Turn      | High       | 11.64120, 78.12168 | 15 KM/Hr    |
| 18 | Turn      | Medium     | 11.63948, 78.12081 | 30 KM/Hr    |
| 19 | Turn      | Medium     | 11.65703, 78.20465 | 30 KM/Hr    |
| 20 | Turn      | High       | 11.58562, 78.74231 | 15 KM/Hr    |
| 21 | Turn      | High       | 11.58528, 78.74234 | 15 KM/Hr    |
| 22 | Turn      | High       | 11.58553, 78.74952 | 15 KM/Hr    |

## Emergency Locations

|   | type     | name                    | coordinates            | speed_limit | risk_level |
|---|----------|-------------------------|------------------------|-------------|------------|
| 0 | hospital | Thilagam Hospital       | 11.4753573, 77.8685829 | 30 km/h     | Medium     |
| 1 | clinic   | Meyyazhagan Clinic      | 11.4752254, 77.8685671 | 30 km/h     | Medium     |
| 2 | hospital | Dr. Jaganathan Hospital | 11.4752717, 77.8709312 | 30 km/h     | Medium     |
| 3 | hospital | Kathir Eye Hospital     | 11.4740976, 77.8681591 | 30 km/h     | Medium     |
| 4 | police   | Police Quarters         | 11.4741243, 77.8673495 | 30 km/h     | Medium     |
| 5 | hospital | Sankari Hospital        | 11.4777496, 77.873892  | 30 km/h     | Medium     |

|    | type     | name                                                            | coordinates               | speed_limit | risk_level |
|----|----------|-----------------------------------------------------------------|---------------------------|-------------|------------|
| 7  | clinic   | Dr. Dhanasekar Clinic                                           | 11.4793113,<br>77.8829247 | 30 km/h     | Medium     |
| 8  | hospital | Sri Vellaiyan Hospital                                          | 11.4798876,<br>77.8831394 | 30 km/h     | Medium     |
| 10 | hospital | C. N. Hospital                                                  | 11.55724,<br>78.0096      | 30 km/h     | Medium     |
| 11 | hospital | Vinayaka Mission's Homeopathic Medical College & Hospital       | 11.583617,<br>78.059145   | 30 km/h     | Medium     |
| 12 | hospital | Vinayaka mission Kirubananadah variyar medical college hospital | 11.584984,<br>78.0618417  | 30 km/h     | Medium     |
| 13 | hospital | vims                                                            | 11.5854239,<br>78.0639188 | 30 km/h     | Medium     |
| 16 | hospital | Vinayaka Mission’s Kirupananda Variyar Medical College          | 11.5977669,<br>78.0850992 | 30 km/h     | Medium     |
| 20 | hospital | Priyam Speciality Hospital                                      | 11.63446,<br>78.124388    | 30 km/h     | Medium     |
| 22 | hospital | Universal Cancer Hospital                                       | 11.6336739,<br>78.1267767 | 30 km/h     | Medium     |
| 23 | hospital | AVM Hospital                                                    | 11.6264369,<br>78.1394204 | 30 km/h     | Medium     |
| 24 | hospital | Dharan Hospital                                                 | 11.6252302,<br>78.1451753 | 30 km/h     | Medium     |
| 25 | hospital | Udhayam Hospital                                                | 11.625819,<br>78.145958   | 30 km/h     | Medium     |
| 26 | pharmacy | மருந்தகம்                                                       | 11.6251501,<br>78.1470098 | 30 km/h     | Medium     |
| 27 | hospital | Rps Hospital                                                    | 11.6260528,<br>78.1474921 | 30 km/h     | Medium     |
| 28 | hospital | மருத்துவமனை நிலம்                                               | 11.625344,<br>78.146058   | 30 km/h     | Medium     |
| 29 | hospital | Akitha Hospital                                                 | 11.6256027,<br>78.1489875 | 30 km/h     | Medium     |
| 30 | hospital | Deepam Speciality Hospital                                      | 11.6269734,<br>78.1476357 | 30 km/h     | Medium     |
| 31 | hospital | Manima Clinic                                                   | 11.623248,<br>78.1465721  | 30 km/h     | Medium     |
| 35 | hospital | Jaaidev Hospital                                                | 11.6655205,<br>78.2634736 | 30 km/h     | Medium     |
| 37 | clinic   | Lavanya Clinic                                                  | 11.652121,<br>78.4795146  | 30 km/h     | Medium     |

|    | type     | name                            | coordinates               | speed_limit | risk_level |
|----|----------|---------------------------------|---------------------------|-------------|------------|
| 38 | hospital | Amsb Agathiyar Sidha Hospital   | 11.650564,<br>78.483542   | 30 km/h     | Medium     |
| 39 | clinic   | Sudha Polyclinic                | 11.647825,<br>78.508537   | 30 km/h     | Medium     |
| 40 | hospital | Government Hospital, Kothampadi | 11.62641,<br>78.542199    | 30 km/h     | Medium     |
| 45 | clinic   | Sugam Clinic                    | 11.5832481,<br>78.7496438 | 30 km/h     | Medium     |
| 46 | hospital | Selvam Hospital                 | 11.585978,<br>78.748761   | 30 km/h     | Medium     |
| 47 | hospital | Prabushankar Hospital           | 11.584239,<br>78.7490323  | 30 km/h     | Medium     |

## Crowded Spots

|    | type        | name                                                       | coordinates               | speed_limit | risk_level |
|----|-------------|------------------------------------------------------------|---------------------------|-------------|------------|
| 6  | marketplace | Sunday Market                                              | 11.4796502,<br>77.8708458 | 30 km/h     | Medium     |
| 14 | university  | Vinayaka Missions Vinayaka Missions<br>Research Foundation | 11.5974056,<br>78.0843617 | 30 km/h     | Medium     |
| 15 | school      | Vinayaka Missions Annapoorani Matric<br>School             | 11.5971662,<br>78.0839876 | 30 km/h     | Medium     |
| 17 | school      | Dist Education Training                                    | 11.6133799,<br>78.1022409 | 30 km/h     | Medium     |
| 18 | school      | DIET - Salem                                               | 11.6144158,<br>78.1025078 | 30 km/h     | Medium     |
| 19 | school      | Diamond Rays Matric HR Sec School                          | 11.6385629,<br>78.1206259 | 30 km/h     | Medium     |
| 21 | school      | Sowdeshwari College                                        | 11.6330643,<br>78.1276021 | 30 km/h     | Medium     |
| 32 | school      | Ganesh College                                             | 11.6566575,<br>78.1991121 | 30 km/h     | Medium     |
| 33 | school      | Thangam ZEE public school                                  | 11.6653353,<br>78.251641  | 30 km/h     | Medium     |
| 34 | college     | AVS College od Arts and Science,<br>Salem.                 | 11.6642672,<br>78.2584134 | 30 km/h     | Medium     |



|    | type    | name                              | coordinates               | speed_limit | risk_level |
|----|---------|-----------------------------------|---------------------------|-------------|------------|
| 36 | college | Mahendra Engineering College      | 11.6672537,<br>78.2641303 | 30 km/h     | Medium     |
| 41 | school  | Vani Vikas Matric Hr.sec school   | 11.6212713,<br>78.5462658 | 30 km/h     | Medium     |
| 42 | school  | Saraswati Metric Hr sec School    | 11.5910827,<br>78.6402264 | 30 km/h     | Medium     |
| 43 | college | Maruthi Polytechnic College       | 11.5856852,<br>78.7201072 | 30 km/h     | Medium     |
| 44 | college | Bharathiyar Educational Institure | 11.5863463,<br>78.7364793 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Roundabout  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.63303, 78.12713

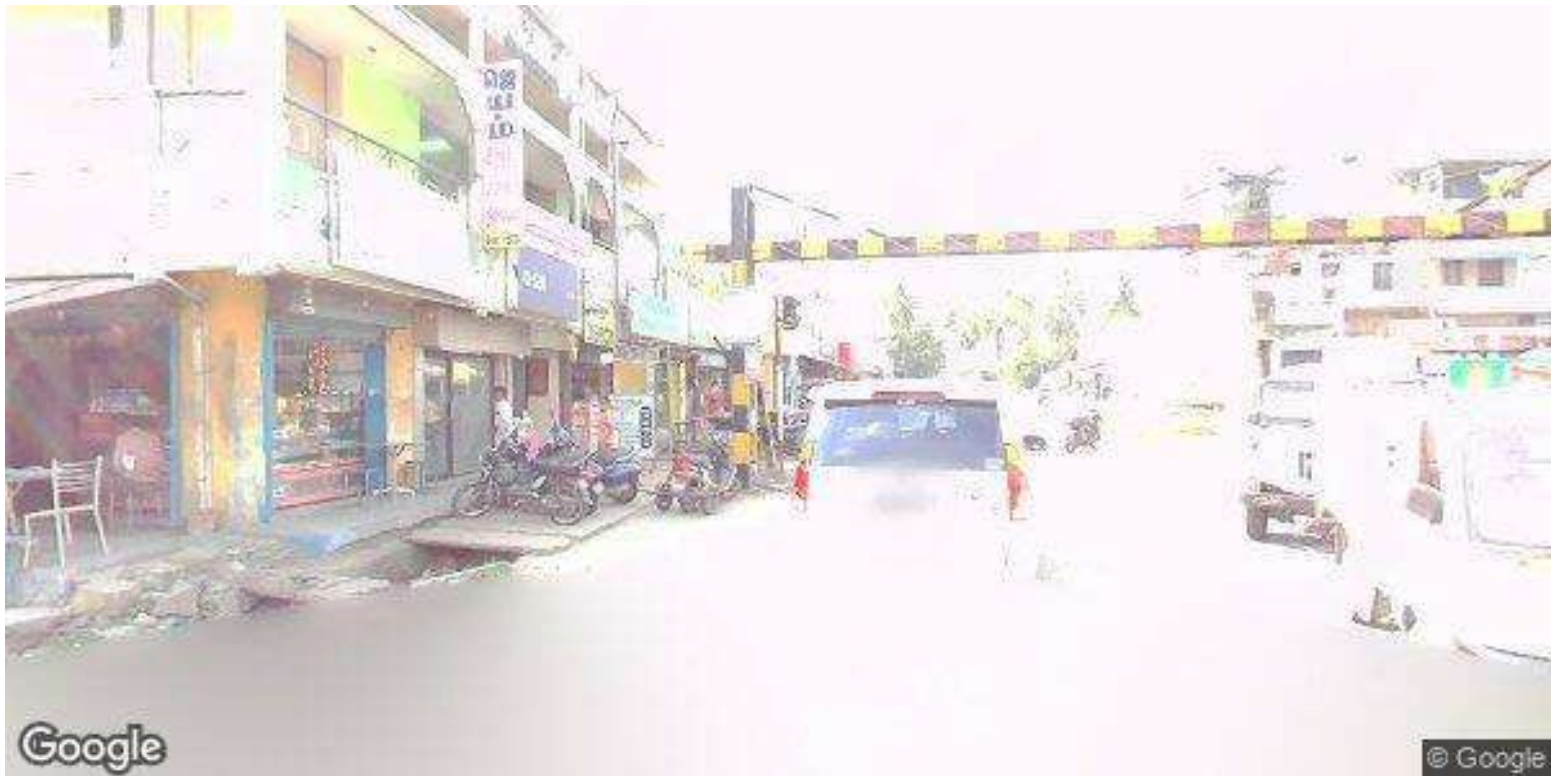




**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44900, 77.87406



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45000, 77.87213





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45028, 77.87202



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45309, 77.87325





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46400, 77.87323



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46765, 77.87037



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.47403, 77.86841





Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.49149, 77.89601



Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.49224, 77.89601





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.53501, 77.95707**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.53601, 77.95856**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.64172, 78.11817**





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.64216, 78.11875**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.64195, 78.12129**





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.64120, 78.12168



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.63948, 78.12081



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**Coordinates:** 11.65703, 78.20465





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.58562, 78.74231



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.58528, 78.74234





Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58553, 78.74952

